

Nonnegative measures belonging to $H^{-1}(\mathbb{R}^2)$.

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Abstract

Radon measures belonging to the negative Sobolev space $H^{-1}(\mathbb{R}^2)$ are important from the point of view of fluid mechanics as they model vorticity of vortex-sheet solutions of incompressible Euler equations. In this note we discuss regularity conditions sufficient for nonnegative Radon measures supported on a line to be in $H^{-1}(\mathbb{R}^2)$. Applying the obtained results, we derive consequences for measures on $\mathbb{R}^2$ with arbitrary support and prove elementarily, among other things, that measures belonging to $H^{-1}(\mathbb{R}^2)$ may be supported on a set of Hausdorff dimension 0. We comment on possible numerical applications.

Keywords: embeddings of measures, vorticity, Hausdorff dimension

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1 Introduction

Let $\mathcal{M}_+(\mathbb{R}^2)$ denote the space of nonnegative bounded Radon measures on $\mathbb{R}^2$ (see [7]) and let $H^{-1}(\mathbb{R}^2)$ be the space of all tempered distributions f on $\mathbb{R}^2$ such that

$$\int_{\mathbb{R}^2} (1 + |y|^2)^{-1} |\hat{f}(y)|^2 dy < \infty.$$

Alternatively, $H^{-1}(\mathbb{R}^2)$ can be viewed as the space of all continuous functionals on the Sobolev space $W^{1,2}(\mathbb{R}^2)$ (see e.g. [1]). The following basic problem can be posed:

Problem A. *Characterize the space $\mathcal{M}_+(\mathbb{R}^2) \cap H^{-1}(\mathbb{R}^2)$.*

Our motivation to study this problem originates in fluid mechanics. Namely, let $u : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the velocity field of a fluid in two-dimensional space and let

$$\omega = \operatorname{curl}(u) := \partial_{x_1} u_2 - \partial_{x_2} u_1$$

be its vorticity field. Then $\omega \in \mathcal{M}_+(\mathbb{R}^2) \cap H^{-1}(\mathbb{R}^2)$ for compactly supported ω means that

- vorticity of the flow is everywhere nonnegative (condition $\omega \in \mathcal{M}_+(\mathbb{R}^2)$),
- kinetic energy of the fluid is locally finite, i.e. $\int_{\Omega} u^2(x) dx < \infty$ for every bounded $\Omega \subset \mathbb{R}^2$ (condition $\omega \in H^{-1}(\mathbb{R}^2)$).

The latter condition follows from the fact that the Biot-Savart operator mapping ω to u by the convolution formula

$$u = K * \omega$$

for $K(x) = \frac{x^\perp}{2\pi|x|^2}$ is bounded from H^{-1} to L^2_{loc} , see below.

Solutions of the incompressible Euler equations,

$$\begin{aligned} \partial_t u + u \nabla u + \nabla p &= 0, \\ \operatorname{div}(u) &= 0. \end{aligned}$$

with vorticity belonging to $\mathcal{M}_+(\mathbb{R}^2)$ were defined and studied in [5]. In [4] Delort proved a basic existence theorem, which states that for initial data $u(t=0, x)$ such that $\omega(0, x) := \operatorname{curl}(u(0, x))$ is a bounded nonnegative Radon measure belonging to $H^{-1}(\mathbb{R}^2)$ there exists a global solution $u(t, x)$ of the Euler equations such that $\omega(t, x) := \operatorname{curl}(u(t, x))$ is a bounded nonnegative Radon measure belonging to $H^{-1}(\mathbb{R}^2)$ for every $t > 0$. Uniqueness of such solutions is still an outstanding open problem. To approach it, it seems reasonable to study Problem A, see also the introduction in [3] for a more comprehensive physical background and motivations.

In the case of compactly supported measures Problem A can be solved as follows. Define the positive logarithmic energy of a measure $\omega \in \mathcal{M}_+(\mathbb{R}^2)$ by

$$\mathcal{H}^+(\omega) := \int_{\mathbb{R}^2} \int_{\mathbb{R}^2} \log^+ \frac{1}{|x-y|} \omega(dx) \omega(dy), \quad (1)$$

where $\log^+(x) = \max(\log(x), 0)$. In [11], which builds upon previous ideas of Delort [4] the following crucial characterization was demonstrated.

Lemma 1.1 (Lemma 3.1 in [11]). *Let ω be a nonnegative measure of finite mass and compact support, and let $u = K * \omega$ be the velocity corresponding to the vorticity ω . Then the following are equivalent:*

1. ω is in H^{-1} .
2. u is in L^2_{loc} .
3. $\mathcal{H}^+(\omega) < \infty$.

As a simple corollary, we obtain that measures belonging to H^{-1} have no discrete part. Indeed, $\mathcal{H}^+(\delta_x) = +\infty$ for every $x \in \mathbb{R}^2$, where δ_x is the Dirac mass in x . For general measures, however, Formula (1) is not very convenient to use and we would like